viii. **Hybrid sterility** – In this type, hybrids are formed due to inter specific crosses but they are sterile due to the failure of the chromosomes to segregate normally during meiosis, example Mule (inter specific cross between a horse and a donkey).

ix. **Hybrid breakdown** – F1 Hybrids are viable and fertile, but F2 hybrids may be inviable or sterile.

6.10 Speciation

The process by which one species evolves into one or more different species is called speciation. **A.E. Emerson** defines species as a 'genetically distinctive, reproductively isolated natural population'. Speciation is a fundamental process in evolution. Evolution of a new species in a single lineage is called an agenesis / phyletic speciation. If one species diverges to become two or more species it is cladogenesis or divergent evolution.

6.10.1 Sympatric speciation/ Reproductive isolation

It is a mode of speciation through which new species form from a single ancestral species while both species continue to inhabit the same geographical region. Two or more species are involved. New species formed due to genetic modification in the ancestor that is naturally selected can no longer breed with the parent population. Sexual isolation is strongest. Phenotypic plasticity has emerged as potentially important first step in speciation initiated within an isolated population.

Phenotypic plasicity is the ability of single genotype to produce more than one phenotype. When this plasticity is expressed seasonally in planktons, it is referred to as cyclomorphosis.

6.10.2 Allopatric speciation/ Geographical speciation

It is a mode of speciation that occurs when biological populations of similar species become isolated from each other that prevents gene flow. One species becomes two species due to geographical barriers hence new species is evolved e.g. Darwin's finches. The barriers are land separation, migration or mountain formation. When barriers occur between species, change in ecological conditions and environment leads to adaptations that produce differences. If there are no adaptations, they will not survive. Sexual isolation is weakest.

A well studied example is the adaptation of Apple maggots that feed on apples in North America. When the apple trees were imported to North America, Apple maggot flies (*Rhagoletis pomonella*) a parasitic insect that normally laid its eggs in the fruit of wild hawthorns until one subset of population began to lay its eggs in the fruit of domesticated apple trees (*Malus domestica*) that grew in the same area. This small group of apple maggot flies selected a different host species from the rest of the population and its offsprings became accustomed to domesticated apples.

6.11 Extinction of Animals

Extinction

Extinction was common if not inevitable because species could not always adapt to large or rapid environmental changes. The impact of extinction can conveniently be considered at three levels.

Species extinction eliminates an entire species, by an environmental event (flood etc.,) or by biological event (disease or non availability of food resource half or more).

Mass extinction eliminates half or more species in a region or ecosystem, as might occur following a volcanic eruption. Five major mass extinction that occurred since the Cambrian period. This mass extinction is often referred to as K-T extinction. **Table 6.2** represents the K-T extinction.

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